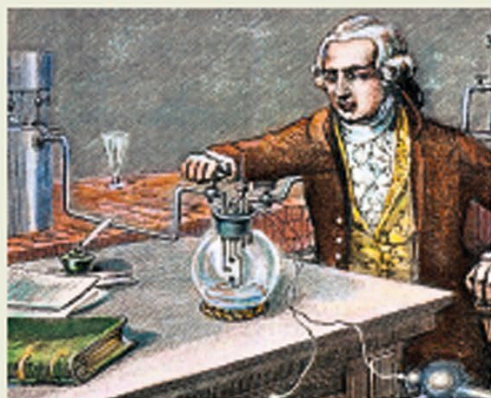
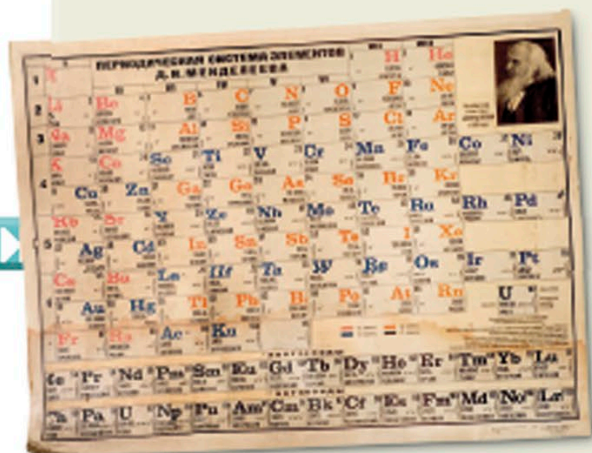




Early attempts

In 1789, French chemist Antoine Lavoisier published his *Elementary Treatise of Chemistry*, in which he grouped 33 chemical elements simply into four types: gases, metals, nonmetals, and earths. Some of his elements were later shown to be compounds (made of two or more elements), such as aluminum oxide.



Mendeleev's periodic table

Russian chemist Dmitri Mendeleev ordered the 59 elements known at the time into eight groups (right) based on their relative atomic mass. His periodic table left three gaps for undiscovered elements, which turned out to be gallium (discovered in 1875), scandium (discovered in 1879), and germanium (discovered in 1886).

Key events

1669

Phosphorus was isolated from urine by German alchemist Hennig Brand (see p.83). It was the first element to be discovered using chemistry.



Phosphorus match tips

1773

English scientist Joseph Priestley isolated oxygen gas, calling it "dephlogisticated air." German chemist Carl Scheele claimed to have also done so, but in 1772.

1829

German chemist Johann Wolfgang Döbereiner organized elements into groups of three with similar chemical properties (such as chlorine, bromine, and iodine). He called these groups triads.

1869

Russian chemist Dmitri Mendeleev published his pioneering Periodic Table in a Russian journal, which was later translated into German and English.

Periodic table

In this color-coded table, chemical elements are arranged in order of their increasing atomic number (the number of protons an atom of an element contains) and in rows, called periods.

All elements in a certain period have the same number of electron shells (the layers of electrons around an atom's nucleus).

In addition, the elements are organized into columns called groups. Each group contains elements with similar chemical properties.

1

1 1.0079

H

Hydrogen

2

3 6.941

Li

Lithium

4 9.0122

Be

Beryllium

11 22.990

Na

Sodium

12 24.305

Mg

Magnesium

19 39.098

K

Potassium

20 40.078

Ca

Calcium

21 44.956

Sc

Scandium

22 47.867

Ti

Titanium

23 50.942

V

Vanadium

24 51.996

Cr

Chromium

25 54.938

Mn

Manganese

37 85.468

Rb

Rubidium

38 87.62

Sr

Strontium

39 88.906

Y

Yttrium

40 91.224

Zr

Zirconium

41 92.906

Nb

Niobium

42 95.94

Mo

Molybdenum

43 (96)

Tc

Technetium

55 132.91

Cs

Caesium

56 137.33

Ba

Barium

57-71

La-Lu

Lanthanides

72 178.49

Hf

Hafnium

73 180.95

Ta

Tantalum

74 183.84

W

Tungsten

75 186.21

Re

Rhenium

87 (223)

Fr

Francium

88 (226)

Ra

Radium

89-103

Ac-Lr

Actinides

104 (261)

Rf

Rutherfordium

105 (262)

Db

Dubnium

106 (266)

Sg

Seaborgium

107 (264)

Bh

Bohrium

57 138.91

La

Lanthanum

58 140.12

Ce

Cerium

59 140.91

Pr

Praseodymium

60 144.24

Nd

Neodymium

89 (227)

Ac

Actinium

90 232.04

Th

Thorium

91 231.04

Pa

Protactinium

92 238.03

U

Uranium

Key

- Hydrogen
- Alkali metals
- Alkaline earth metals
- Transition metals
- Lanthanide series
- Actinide series
- Other metals
- Metalloids
- Other nonmetals
- Halogens
- Noble gases

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